

Descriptive analysis of harmonized broadband data

This document provides a brief descriptive overview of the harmonized German municipal broadband panel dataset (2005-2021).

Last updated: July 12, 2026 (v3.0.0). Plots were regenerated from `output/panel_data_public.csv`.

Data quality notes

Use with caution

This dataset has known limitations that users should be aware of:

1. **Methodological breaks:** the 2010 baseline definition change, tier changes at 2013 and 2018, and the 2015 provider change affect comparability across years (see below)
2. **Missing years and tiers:** 2009 is absent from the municipality panel; 2005-2008 only report the historical baseline tier; the 1, 6, and 30 Mbps tiers are only reported from 2018, so before then the higher-tier share columns are measured at stricter tiers (recorded in the `tier_*` columns)
3. **2010-2014 covers only reporting municipalities** (~4,400-4,500 per year, skewed toward larger municipalities); non-reporting municipalities coded as zero were removed in v2.0.0
4. **Trailing non-reporting zeros removed (2015-2021):** 141 municipality-years, mostly gemeindefreie Gebiete dropping from full coverage to 0% at 2021, were removed in v3.0.0 (`output/nonreporting_zero_blocks_2015_2021.csv`); persistent and leading zeros are retained
5. **Filtered observations:** Step 06 filters 9,726 post-duplicated long rows (0.32%) covering 121 historical AGS codes that cannot be mapped to Destatis 2021 boundaries (see `unmapped_ags_documentation.md` for details)

If using prior data

If you previously used this dataset, please re-download and see `CHANGELOG.md`. Highlights: v2.0.0 removed ~33,000 false-zero municipality-years in 2010-2014, added Berlin and Hamburg, and recovered previously unmapped municipalities; v2.1.0 added the `tier_*` columns that record the speed tier behind each share value and corrected codebook errors; v3.0.0 removed 141 trailing non-reporting zeros in 2015-2021 and recovered one further unit.

Discontinuous changes in the data

The panel data is constructed from historical sources with changing methodologies and changing speed-tier menus. This results in several breaks in the time series that users must be aware of.

The 2009/2010 break

To create a continuous series for basic internet availability, the `share_broadband_baseline` variable was constructed from different underlying sources:

- **For 2005-2008:** The variable is based on historical DSL availability data (`speed_mbps_gte = 0.128`, representing ≥ 0.128 Mbps).
- **For 2009:** No municipality panel is available in the current input files.
- **From 2010 onwards:** The variable is based on the lowest reported tier: ≥ 2 Mbps in 2010-2017 and ≥ 1 Mbps from 2018 (see `tier_baseline`). Since these imply ≥ 0.128 Mbps, they are lower bounds on basic access.

The consequence is a discontinuity in the data where the definition of the baseline metric changes. In the current panel, this appears as a drop between 2008 and 2010, when the source switches from historical DSL availability to the stricter tier. This is distinct from organic change in broadband availability.

The 2013 and 2018 tier breaks

The share columns take the lowest reported tier at or above their named threshold, and the reported tiers change over time. `share_gte6mbps` is measured at ≥ 50 Mbps in 2010-2012 (where it is identical to `share_gte30mbps` by construction), at ≥ 16 Mbps in 2013-2017, and at ≥ 6 Mbps from 2018. `share_gte30mbps` is measured at ≥ 50 Mbps through 2017 and at ≥ 30 Mbps from 2018. `share_gte1mbps` and the baseline are measured at ≥ 2 Mbps through 2017 and ≥ 1 Mbps from 2018. Part of the level jumps at 2013 and 2018 is therefore definitional, not rollout. The per-observation `tier_*` columns identify the measurement tier; analyses spanning these years should condition on them or restrict to constant-tier windows.

The 2015 break

2015 marks a change in the primary data provider and reporting standards. Since the removal of the 2010-2014 false zeros in v2.0.0, the remaining content of this break is mostly a composition change: the panel returns to near-complete municipality coverage in 2015, while the within-municipality baseline jump among continuous reporters is only 1.2 percentage points. The `method_change_2015` dummy variable is included in the dataset to allow researchers to control for this event.

Descriptive plots

Average annual coverage

This plot shows the mean coverage for the baseline metric alongside higher speed tiers. The dashed lines mark the **2010 definition break**, the **2013 and 2018 tier breaks**, and the **2015 method break**. The sharp rises in the ≥ 6 Mbps series at 2013 and 2018 sit exactly on the tier breaks and are partly definitional.

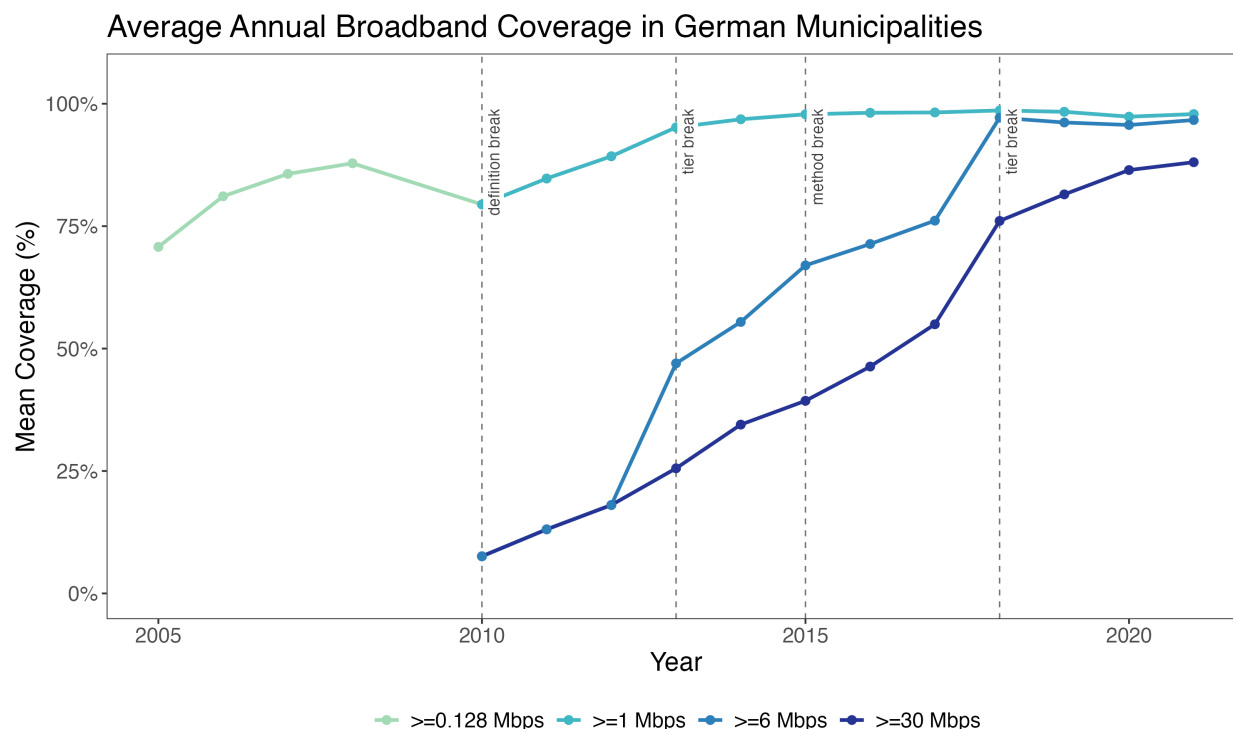


Figure 1: Average Annual Coverage

Year-over-year change in coverage

This plot highlights the magnitude of the year-on-year changes in unweighted means. Changes at the dashed break lines mix definitional shifts and sample-composition changes with organic rollout; changes elsewhere reflect rollout among reporting municipalities.

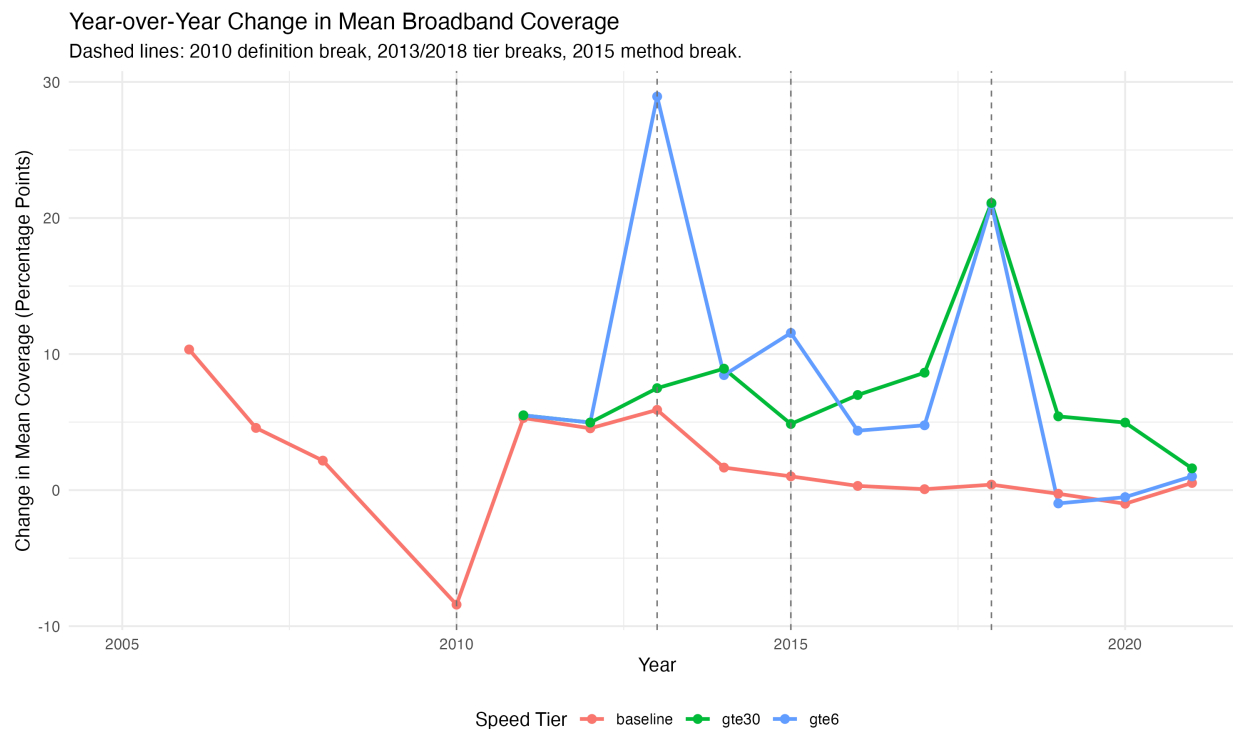


Figure 2: Year-over-Year Change in Coverage

Distribution of coverage levels

This set of histograms shows the distribution of coverage values. The `share_broadband_baseline` is heavily skewed towards 100%, indicating that most municipalities achieved full basic coverage relatively early. In contrast, higher speed tiers show more variation and a larger concentration near 0%.

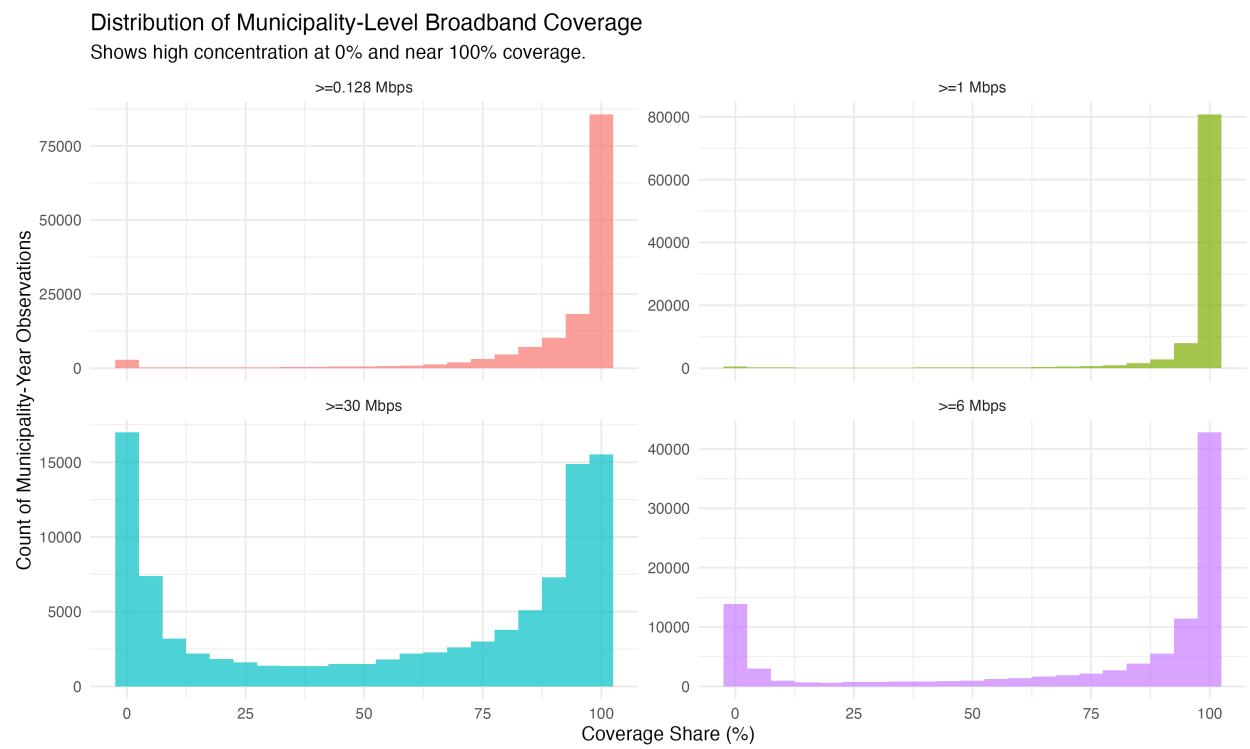


Figure 3: Distribution of Coverage Levels